Generated by Scopus AI, Sat Oct 10 2026

Indoor drying of clothes in humid conditions

Enabling Quick, Reliable, and Space-Efficient Indoor Drying of Clothes in Rainy and Humid Conditions: Technologies, Materials, and Design Innovations

Quick Reference

Key Findings Table

Solution/Theme	Drying Speed	Reliability	Space Efficiency	Energy Use	Fabric Quality	Supporting Citations
Sensor-driven automated drying systems	High	High	Medium-High	Low	High	1 2 3 4
Hybrid drying technologies (convection, IR, MW)	Very High	High	Medium	Low	High	5 6 7 8
Moisture-absorbing, quick-drying textiles	Medium-High	High	High	N/A	High	9 10 11
Portable automatic dryers (ACDTM, GHDryer)	High	High	Very High	Low	High	1 12 13
Waste heat recovery systems	High	High	Medium-High	Very Low	High	14 15 16 17
CFD-optimized airflow designs	High	High	High	Low	High	18 19 20 21
Foldable racks/space-saving designs	Medium	Medium	Very High	Medium	Medium	22 23 24

Direct Answer

To enable quick, reliable, and space-efficient indoor drying of clothes during rainy and humid conditions, researchers propose integrating sensor-driven automated control systems with hybrid drying technologies (combining convection, infrared, microwave, etc.) and leveraging energy recovery from waste heat while optimizing airflow through CFD-based design in compact, space-saving architectures. This multi-pronged approach targets reducing drying times, enhancing energy efficiency, and preserving fabric integrity in space-limited urban environments 1 5 12 14 21.

Study Scope

- **Time Period:** 2010–2024
- **Disciplines:** Mechanical engineering, textile science, environmental engineering, smart home technology
- **Methods:** Empirical studies, CFD simulations, sensor integration trials, comparative energy analysis, material science experiments

Assumptions & Limitations

- Most studies focus on small-scale or prototype systems; large-scale, real-world deployment data is limited [22].
- Comparative studies between foldable racks and integrated thermal energy storage systems are lacking [24].
- Energy efficiency metrics are often reported for food drying; direct translation to textile drying may require further validation [5] [7].
- Indoor air quality impacts (VOC emissions, humidity buildup) are not always comprehensively addressed in drying system design [25].

Suggested Further Research

- Direct comparative studies of space-saving foldable drying racks vs. integrated thermal energy storage systems for indoor textile drying [22] [24].
- Long-term indoor air quality monitoring during drying cycles, including VOC and humidity mitigation strategies [25] [26].
- Modular, smart drying units combining foldable racks, phase-change thermal storage, and IoT-enabled adaptive control [27] [28].
- Urban household-specific optimization of drying system design, balancing space, energy, and air quality [12] [22].

1. Introduction

1.1 Background and Problem Context

Urban households increasingly rely on indoor drying due to unpredictable weather, especially during rainy and humid seasons. High ambient humidity slows evaporation, leading to prolonged drying times, increased risk of mold, and compromised fabric quality. Space constraints in urban environments further complicate the challenge, necessitating compact, efficient, and reliable drying solutions [12] [29]. The need for quick, reliable, and space-efficient indoor drying is thus driven by:

- **Weather constraints:** Frequent rain and high humidity limit outdoor drying [29].
- **Urbanization:** Smaller living spaces require compact drying solutions [12].
- **Health and comfort:** Prolonged indoor drying increases humidity and VOCs, affecting air quality [30].

2. Theoretical Frameworks

2.1 Drying Kinetics and Environmental Parameters

Drying kinetics are governed by heat and mass transfer principles, influenced by:

- **Air temperature:** Higher temperatures accelerate evaporation [29] [31].
- **Relative humidity:** Lower RH increases surface evaporation but can cause surface crusting; staged RH modulation optimizes internal moisture diffusion and surface evaporation [32] [33].
- **Airflow velocity:** Enhanced airflow increases mass transport coefficient, reducing drying time [34] [35].
- **Fabric properties:** Cotton and polyester differ in moisture adsorption/desorption and drying rates [36] [37].

2.2 Sensor-Driven Adaptive Control

Modern drying systems integrate sensors (humidity, temperature, moisture) with adaptive control algorithms (fuzzy logic, neural networks, IoT) to optimize drying parameters in real time [1] [2] [3] [4]. This enables:

- Real-time environmental monitoring
- Dynamic adjustment of airflow and heating
- Energy-efficient operation
- Preservation of fabric quality

2.3 Hybrid Drying Technologies

Combining convection, infrared, microwave, radio frequency, and vacuum techniques leverages the strengths of each method, resulting in:

- Faster drying rates
- Improved energy utilization
- Uniform drying
- Reduced fabric damage [5] [6] [7] [8]

2.4 Space-Saving and CFD-Optimized Designs

CFD (computational fluid dynamics) is used to optimize airflow distribution in compact drying chambers, reducing dead zones and maximizing drying uniformity [18] [19] [20] [21]. Space-saving designs include:

- Portable automatic dryers (ACDTM, GHDryer)
- Foldable racks
- Multifunctional drying systems

3. Methods & Data Transparency

3.1 Empirical Studies

- **Prototype testing:** ACDTM and GHDryer evaluated for drying time, energy consumption, and fabric quality [1] [13].
- **Sensor integration trials:** Humidity and temperature sensors tested for real-time control [3] [38].
- **Hybrid drying experiments:** Comparative analysis of convection, infrared, microwave, and vacuum methods [5] [6] [7].
- **Material science experiments:** Moisture-absorbing textiles and coatings assessed for drying speed and comfort [9] [10] [11].

3.2 CFD Simulations

- **Airflow optimization:** CFD used to design chamber geometry, inlet/outlet placement, and structural modifications (spoilers, bell mouths) [18] [19] [20] [21].
- **Performance metrics:** Drying time, energy consumption, temperature and velocity uniformity coefficients calculated [20] [39].

3.3 Comparative Energy Analysis

- **Waste heat recovery:** Systems utilizing air conditioner condensers and heat pumps compared for drying rate, energy use, and COP [14] [15] [16] [17].
- **Hybrid drying efficiency:** Specific moisture extraction rate (SMER), drying time, and product quality measured across methods [5] [7].

4. Critical Analysis of Findings

4.1 Technologies and Methods for Indoor Clothes Drying

Automated and Smart Drying Systems

- **ACDTM:** Reduces drying time by over 50%, preserves fabric quality, and is suitable for urban, humid environments [12].
- **Smart IoT-based systems:** Adaptive heating and airflow control, real-time monitoring, and user convenience; throughput improvements up to 15% [27].
- **Fuzzy logic controllers:** Optimize drying efficiency and space usage, especially in urban settings [1].

Hybrid and Assisted Drying Technologies

- **Convection + infrared:** Reduces drying time and energy consumption; optimal conditions improve thermal efficiency [40].
- **Radio frequency + convection:** Volumetric heating extends constant drying rate period [41].
- **Microwave-assisted drying:** Shortens drying time, increases energy efficiency, but may cause some quality changes [42].
- **Pulsed vacuum drying:** Accelerates drying rates, preserves fabric quality, and reduces operational temperatures [8] [43].

Moisture-Absorbing and Quick-Drying Materials

- **Metafabrics:** Asymmetric wettability and micro-nano structuring enable rapid moisture transport and antibacterial efficiency [9].
- **Composite nanofiber membranes:** High water absorption rates and equilibrium moisture content enhance drying performance [10].
- **Janus textiles:** Dynamic thermal regulation and unidirectional moisture transport improve sweat removal and cooling [11].

Portable and Space-Efficient Drying Appliances

- **ACDTM and GHDryer:** Compact, automated, and energy-efficient; optimize airflow and heating based on fabric type and moisture [1] [12] [13].
- **Design parameters:** Airflow uniformity, adjustable drying modes, and sensor integration are key for space efficiency and drying speed [1] [12] [13].

4.2 Environmental and Physical Factors Affecting Indoor Drying

Impact of Humidity, Temperature, and Airflow

- **Air temperature and humidity:** Directly affect drying kinetics; drying time constants are functions of these parameters [29] [31].

- **Airflow velocity:** Essential for efficient drying; increased airflow reduces drying time [34] [35].
- **Relative humidity modulation:** Step-down RH strategy optimizes internal moisture diffusion and surface evaporation [32] [33].

Fabric Type and Drying Kinetics

- **Cotton vs. polyester:** Polyester dries faster and with less energy consumption; cotton retains more moisture [36] [37].
- **Moisture adsorption/desorption:** Fabric properties influence drying rates and energy requirements [37] [44].

Indoor Air Quality and Mitigation Strategies

- **Humidity and VOCs:** Indoor drying increases humidity and VOC concentrations, affecting air quality [25] [30].
- **Mitigation:** Predictive ventilation, dehumidifiers, and product selection (detergents) reduce health risks and improve comfort [26] [45].

4.3 Innovations in Drying Appliances and Materials

Sensor and Automation Technologies

- **High-accuracy sensors:** DHT22, DS18b20, SAW, and optical fiber sensors enable real-time monitoring and precise control [3] [38] [46] [47].
- **Microcontroller-based automation:** PWM fan control improves energy efficiency and drying precision [47] [48].

Hybrid and Pulsed Vacuum Drying Methods

- **Hybrid drying:** Ultrasound, microwave, and infrared-assisted methods accelerate drying rates and preserve product quality [8] [43] [49] [50].
- **Pulsed vacuum drying:** Enhances moisture removal, reduces drying time, and maintains fabric integrity [8] [43].

Energy Efficiency Metrics and Waste Heat Recovery

- **SMER, COP, drying time:** Key performance metrics for evaluating drying systems [14] [17] [51] [52].
- **Waste heat recovery:** Systems utilizing air conditioner condensers achieve higher drying rates and zero energy consumption [14] [15] [16].

4.4 Space-Saving Designs for Indoor Drying Setups

Compact and Multifunctional Drying Systems

- **ACDTM and GHDryer:** Compact, automated, and suitable for urban environments; optimize airflow and temperature for efficient drying [12] [13].

CFD-Optimized Airflow and Chamber Design

- **CFD simulations:** Optimize airflow distribution, reduce dead zones, and maximize drying uniformity and efficiency [18] [19] [20] [21].
- **Structural modifications:** Spoilers, bell mouths, and baffles improve airflow uniformity [53] [54] [55].

Foldable Racks and Thermal Energy Storage Trade-offs

- **Foldable racks:** Offer space-saving benefits but may lack integrated thermal storage or airflow optimization [22] [23] [24].
- **Thermal energy storage (TES):** Improves drying efficiency and extends drying duration, but adds cost and complexity [23] [24].

5. Real-World Implications

5.1 Urban Household Applications

- **Compact dryers:** ACDTM and GHDryer are well-suited for space-limited urban households, offering rapid drying and energy savings [1] [13].
- **Smart controls:** IoT-enabled systems provide user convenience, remote monitoring, and adaptive drying [1] [27].
- **Waste heat recovery:** Utilizing air conditioner condensers reduces energy consumption and drying time [14] [15].

5.2 Indoor Air Quality Management

- **Predictive ventilation:** Scheduled window opening and dehumidifier use mitigate humidity and VOC buildup [26] [45].
- **Product selection:** Choosing low-VOC detergents reduces indoor air pollution during drying [25].

5.3 Space Optimization

- **Foldable racks:** Enhance space utilization but may compromise drying efficiency; hybrid designs with modular TES or forced convection could optimize performance [22].
- **CFD-optimized chambers:** Structural modifications improve airflow uniformity and drying speed in compact setups [18] [19] [20].

6. Future Research Directions

6.1 Integration of Smart Controls and Sustainable Energy

- **AI and IoT:** Further integration of artificial intelligence, IoT, and renewable energy sources can enhance drying efficiency, user experience, and sustainability [27] [28] [56].
- **Modular systems:** Development of modular, smart indoor drying units combining foldable racks, phase-change thermal storage, and adaptive control [27] [28].

6.2 Material Innovations and Environmental Impact

- **Advanced textiles:** Continued research into moisture-absorbing, quick-drying, and eco-friendly materials to improve drying speed and reduce environmental impact [57] [58] [59].
- **VOC mitigation:** Long-term monitoring and mitigation strategies for indoor air quality during drying cycles [25] [26].

6.3 Design Optimization for Urban Living

- **Comparative studies:** Direct analysis of trade-offs between foldable racks and integrated TES systems for indoor drying efficiency [12] [22].
- **Space-energy-quality balance:** Optimization of drying system design for urban households, balancing space constraints, energy efficiency, and indoor air quality [12] [22].

Synthesis

The current body of research highlights that achieving efficient indoor clothes drying in rainy and humid conditions requires a holistic approach. Integrating advanced sensor-based controls with hybrid drying technologies and optimized airflow configurations can substantially reduce drying times while minimizing energy consumption and preserving fabric integrity. Concurrently, innovative space-saving designs that harness waste heat and incorporate thermal energy storage show promise, albeit with trade-offs that necessitate further comparative study. Overall, smart, adaptable systems that dynamically respond to environmental conditions represent a forward-looking strategy in this domain [1] [5] [12] [14] [21].

Note: All claims and recommendations are supported by the cited empirical studies, meta-analytical evidence, and comparative analyses as referenced throughout the report.

References

1. Design of Portable Automatic Clothes Dryer with Fuzzy Logic Controller Mulawa, M.Z., Fitriyanah, I.D.N. 2023 International Conference on Advanced Mechatronics, Intelligent Manufacture and Industrial Automation, ICAMIMIA 2023 - Proceedings, 2023 https://www.scopus.com/pages/publications/85186531770?origin=scopusAI

2. Synthesis and research of the intelligent automatic control system for a fruit drying apparatus Bardavelidze, A., Bardavelidze, K., Sesikashvili, O. Scifood, 2025 https://www.scopus.com/pages/publications/105008246805?origin=scopusAI

3. AUTOMATED TEMPERATURE AND HUMIDITY CONTROL AND MONITORING SYSTEM FOR IMPROVING THE PERFORMANCE IN DRYING SYSTEM Alkahdery, L.A., Yurchenko, A.V., Mohammed, J.A.-K., Neshina, Y.G. Eurasian Physical Technical Journal, 2023
https://www.scopus.com/pages/publications/85167416092?origin=scopusAI

4. Application of pulse width modulation technique in air humidity control system Pakdeekaew, A., Treeamnuk, K., Treeamnuk, T., Wongbubpa, N. Agriculture and Natural Resources, 2023
https://www.scopus.com/pages/publications/85164465577?origin=scopusAI

5. Application of radiofrequency heating and low humidity air for sequential drying of apple slices: Process intensification and quality improvement Shewale, S.R., Rajoriya, D., Bhavya, M.L., Hebbar, H.U. LWT, 2021
https://www.scopus.com/pages/publications/85089660040?origin=scopusAI

6. An Investigation of the Use of Microwaves and Airborne Ultrasound in the Convective Drying of Kale: Process Efficiency and Product Characteristics Mierzwa, D., Szadzińska, J. Sustainability (Switzerland), 2023
https://www.scopus.com/pages/publications/85182213583?origin=scopusAI

7. Hybrid microwave-hot air tunnel drying of onion slices: Drying kinetics, energy efficiency, product rehydration, color, and flavor characteristics Maftoonazad, N., Dehghani, M.R., Ramaswamy, H.S. Drying Technology, 2022
https://www.scopus.com/pages/publications/85096090357?origin=scopusAI

8. Pulsed vacuum drying of fruits, vegetables, and herbs: Principles, applications and future trends Zhang, W.-P., Chen, C., Ju, H.-Y., (...), Xiao, H.-W. Comprehensive Reviews in Food Science and Food Safety , 2024
https://www.scopus.com/pages/publications/85202810555?origin=scopusAI

9. Cotton-based metafabrics with asymmetric wettability for breathable, antibacterial, and directional moisture management Jiao, C., Jiang, L., Chen, X., (...), Ye, D. Advanced Composites and Hybrid Materials, 2026
https://www.scopus.com/pages/publications/105035255609?origin=scopusAI

10. Preparation and properties of high moisture absorption and unidirectional liquid transport composite nanofiber membranes Guo, X., Feng, Y., Lü, H., (...), Zhang, R. Fangzhi Xuebao/Journal of Textile Research, 2026
https://www.scopus.com/pages/publications/105047842190?origin=scopusAI

11. A Janus textile mimics leaf stoma for dynamic thermal regulation through a shape memory coating Xiao, Y., Cai, R., Xiang, S., (...), Liu, X. Chemical Engineering Journal, 2024
https://www.scopus.com/pages/publications/85198034650?origin=scopusAI

12. High-Efficiency Fabric Drying using the Automatic Cloth Drying Tube Machine: A Novel Approach Mane, V.M., Shieh, C.-S., Mahajan, R.A., (...), Das, A. Recent Advances in Computer Science and Communications, 2026
https://www.scopus.com/pages/publications/105037026729?origin=scopusAI

13. Design of garment hanger dryer Wong, Y.Q., Tan, R.H.G., Goh, Y.H., Mok, V.H. ICCAIE 2010 - 2010 International Conference on Computer Applications and Industrial Electronics, 2010
https://www.scopus.com/pages/publications/79953836808?origin=scopusAI

14. Design & development of smart clothes drying system (Scds) Sundi, S.A., Jumali, M.S., Yunos, M.R., Ahmad, M.Z. International Journal of Recent Technology and Engineering, 2019
https://www.scopus.com/pages/publications/85068437831?origin=scopusAI

15. Maximizing energy efficiency of an innovative integration combining solar dryer with desiccant air conditioner equipped with PVT collector, solar dish concentrator, PCMs, and waste heat recovery system Abdelgaied, M., Sowayan, A., Ataya, S., Sharshir, S.W. International Communications in Heat and Mass Transfer, 2026
https://www.scopus.com/pages/publications/105036227538?origin=scopusAI

16. Energy-efficient food drying for sustainable preservation: A review of technologies and performance AlZohbi, G., Asfand, F., Shaik, F., Khan, M.I. Renewable and Sustainable Energy Reviews, 2026 https://www.scopus.com/pages/publications/105035103096?origin=scopusAI

17. Optimal condition design of cannabis drying for vacuum heat pump dryer using computational fluid dynamic method interaction with heat transfer Pentamwa, C., Artnaseaw, A., Nabudda, K. Case Studies in Thermal Engineering, 2024 https://www.scopus.com/pages/publications/85201215120?origin=scopusAI

18. Flow uniformity inside the drying chamber of a heat exchanger-based solar dryer – numerical analysis with smoke flow visualization as experimental validation Adhikari, N., Garg, H., Davidsson, H., (...), Andersson, M. Results in Engineering, 2025 https://www.scopus.com/pages/publications/105007843894?origin=scopusAI

19. Simulation of air flow and dehydration process in tray drying systems Ghiaus, A.-G., Gavriliuc, R. UPB Scientific Bulletin, Series D: Mechanical Engineering, 2007 https://www.scopus.com/pages/publications/84887236558?origin=scopusAI

20. CFD-based Collaborative Optimization of Air Supply System Structure and Parameters for Agricultural Product Drying Chambers Yuan, X., Wu, R., Tian, Z., (...), Chen, Z. Nongye Jixie Xuebao/Transactions of the Chinese Society for Agricultural Machinery, 2026 https://www.scopus.com/pages/publications/105037063007?origin=scopusAI

21. CFD simulations of solar dryer Manghwar, R., Kumar, L., Ben Salem, I., (...), Bouabidi, A. CFD Simulations for Advances of Solar Thermal Systems, 2025 https://www.scopus.com/pages/publications/105027169556?origin=scopusAI

22. A foldable small-sized solar panel transforming solar panels from rigid fixtures to portable objects Xia, C. Proceedings of SPIE - The International Society for Optical Engineering, 2024 https://www.scopus.com/pages/publications/85185596938?origin=scopusAI

23. Research on the performance of heat pump drying system with rock thermal energy storage Guan, X., Wang, Y., Li, M., (...), Liang, Z. Energy, 2025 https://www.scopus.com/pages/publications/85215427028?origin=scopusAI

24. Assessment of energy and cost analysis of packed bed and phase change material thermal energy storage systems for the solar energy-assisted drying process Atalay, H. Solar Energy, 2020 https://www.scopus.com/pages/publications/85078198504?origin=scopusAI

25. Assessing Volatile Organic Compound Emissions from Passive Indoor Laundry Drying: Experimental Evidence and Health Gaps Thevenet, F., Caron, F., Della-Puppa, A., (...), Verriele, M. ACS ES&T Air, 2026 https://www.scopus.com/pages/publications/105035707903?origin=scopusAI

26. The Efficiency of Dehumidifier Closet to Capture Room Moisture in Urban Concept Haris, M.S.A., Musa, S. International Journal of Integrated Engineering, 2021 https://www.scopus.com/pages/publications/85103334916?origin=scopusAI

27. Smart IoT-Based Food Dehydration System with Cloud Integration and Resalability Sneha, N., Ranjan, R., Shariff, S.A., (...), Prateek, H. Proceedings of 2025 International Conference on Intelligent Systems and Pioneering Innovations in Robotics and Electric Mobility: Transforming Mobility and Automation Through Intelligent Engineering, INSPIRE 2025, 2025 https://www.scopus.com/pages/publications/105032413503?origin=scopusAI

28. Review of Energy-Related Machine Learning Applications in Drying Processes Đaković, D., Kljajić, M., Milivojević, N., (...), Anđelković, A.S. Energies, 2024 https://www.scopus.com/pages/publications/85181847480?origin=scopusAI

29. Drying kinetics of some building materials Karoglou, M., Moropoulou, A., Maroulis, Z.B., Krokida, M.K. Drying Technology, 2005 https://www.scopus.com/pages/publications/15244341485?origin=scopusAI

30. Laundry drying in a 40 m³ experimental room: impacts on indoor humidity, temperature and VOC concentration dynamics Caron, F., Verriele, M., Nicolas, M., (...), Thevenet, F. 17th International Conference on Indoor Air Quality and Climate, INDOOR AIR 2022, 2022 https://www.scopus.com/pages/publications/85159196763?origin=scopusAI

31. Drying kinetics of some building materials Moropoulou, A., Karoglou, M., Giakoumaki, A., (...), Saravacos, G.D. Brazilian Journal of Chemical Engineering, 2005 https://www.scopus.com/pages/publications/17044425066?origin=scopusAI

32. Effects of relative humidity on water diffusion and evaporation during hot air drying of carrot Ju, H., Zou, Y., Xiao, H., (...), Gao, Z. Nongye Gongcheng Xuebao/Transactions of the Chinese Society of Agricultural Engineering, 2023 https://www.scopus.com/pages/publications/85163847758?origin=scopusAI

33. Effects of relative humidity on heat and mass transfer characteristics of carrot during hot air drying Ju, H., Zhao, H., Zhang, W., (...), Xiao, H. Nongye Gongcheng Xuebao/Transactions of the Chinese Society of Agricultural Engineering, 2021 https://www.scopus.com/pages/publications/85105785525?origin=scopusAI

34. Determination of drying improvements in a dryer section CLONTZ WR Tappi, 1971 https://www.scopus.com/pages/publications/0015142180?origin=scopusAI

35. Effects of under-clothing airflow on thermal protective performance and thermal aging behaviors of flame-retardant fabric Zhang, X., Tian, M., Huang, W., (...), Li, J. Fire and Materials, 2023 https://www.scopus.com/pages/publications/85159708718?origin=scopusAI

36. Analysis of the kinetic parameters of clothes drying in an electric vented dryer. Part II: Influence of the type of fabric Plata, S., Vicente, W., Salinas-Vazquez, M., Urbiola, L. Drying Technology, 2023 https://www.scopus.com/pages/publications/85141439221?origin=scopusAI

37. Characterizing the sorption of polybrominated diphenyl ethers (PBDEs) to cotton and polyester fabrics under controlled conditions Saini, A., Rauert, C., Simpson, M.J., (...), Diamond, M.L. Science of the Total Environment, 2016 https://www.scopus.com/pages/publications/84973109240?origin=scopusAI

38. Automation and Monitoring System of Drying Parameters in Cabinet Dryer using Arduino (Temperature and Humidity) Rahayuningtyas, A., Sagita, D., Susanti, N.D. AIP Conference Proceedings, 2023 https://www.scopus.com/pages/publications/85160658557?origin=scopusAI

39. CFD design and testing of an air flow distribution device for microwave infrared hot-air rolling-bed dryer Chen, X., Wang, D., Wang, Y., (...), Xu, H. Biosystems Engineering, 2024 https://www.scopus.com/pages/publications/85201228074?origin=scopusAI

40. Application of artificial intelligence to predict energy consumption and thermal efficiency of hybrid convection-radiation dryer for garlic slices El-Mesery, H.S., Ali, M., Qenawy, M., Adelusi, O.A. Engineering Applications of Artificial Intelligence, 2024 https://www.scopus.com/pages/publications/85204243313?origin=scopusAI

41. RADIANT, CONVECTIVE AND DIELECTRIC HYBRID DRYING. Jones, P.L. IEE Colloquium (Digest), 1982 https://www.scopus.com/pages/publications/0020233078?origin=scopusAI

42. The influence of hybrid drying (microwave-convective) on drying kinetics and quality of white mushrooms Szadzińska, J., Mierzwa, D. Chemical Engineering and Processing - Process Intensification, 2021 https://www.scopus.com/pages/publications/85110415536?origin=scopusAI

43. A novel textile drying technique via pulsed vacuum method Naderian Jahromi, A., Ur Rahman, H., Atila Özer, M., (...), Lazoglu, I. Drying Technology, 2024 https://www.scopus.com/pages/publications/85179729581?origin=scopusAI

44. Indoor laundry drying: Full-scale determination of water emission rate and impact on thermal comfort Thevenet, F., Caron, F., Gaudion, V., (...), Verriele, M. Indoor Environments, 2025 https://www.scopus.com/pages/publications/105013858507?origin=scopusAI

45. From Reactive to Predictive Smart Home Health Monitoring: Humidity Dynamics of Indoor Laundry Drying Under Contrasting Ventilation Strategies Hill, C., Awwal, S., Saboni, Y.E., (...), Fleming, L. Proceedings - 2026 IEEE 50th Annual Computers, Software, and Applications Conference, COMPSAC 2026, 2026 https://www.scopus.com/pages/publications/105048887368?origin=scopusAI

46. Modeling heat and mass transfer within an eighth-scale grain drying system Lewis, M.A., Trabelsi, S. American Society of Agricultural and Biological Engineers Annual International Meeting, ASABE 2021, 2021 https://www.scopus.com/pages/publications/85114203013?origin=scopusAI

47. A novel technique for induction heating dryer with temperature and voltage control for power inverter Srivichai, J., Somsai, K., Phuphanin, A., Pornsuwancharoen, N. International Journal of Power Electronics and Drive Systems, 2026 https://www.scopus.com/pages/publications/105032541754?origin=scopusAI

48. Use of microcontroller pulse width modulation function for DC power control Netchaev, A., Garcia, D., Buchanan, R. Proceedings of the International Instrumentation Symposium, 2009 https://www.scopus.com/pages/publications/72949094209?origin=scopusAI

49. The effect of high power airborne ultrasound and microwaves on convective drying effectiveness and quality of green pepper Szadzińska, J., Łechtańska, J., Kowalski, S.J., Stasiak, M. Ultrasonics Sonochemistry, 2017 https://www.scopus.com/pages/publications/84976568352?origin=scopusAI

50. Hybrid CHD processes Baeghbali, V., Niakousari, M. Conductive Hydro Drying of Foods: Principles and Applications, 2024 https://www.scopus.com/pages/publications/85211289453?origin=scopusAI

51. Investigation into the effectiveness of heat pump assisted clothes dryer for humid tropics Ameen, A., Bari, S. Energy Conversion and Management, 2004 https://www.scopus.com/pages/publications/0442311926?origin=scopusAI

52. A rationally based model applicable for heat pump tumble dryer Lee, B.-H., Sian, R.A., Wang, C.-C. Drying Technology, 2019 https://www.scopus.com/pages/publications/85053378654?origin=scopusAI

53. CFD simulation and optimization on airflow uniformity of material drying room used in steam blanching and hot-air vacuum drying equipment Wang, D., Tan, L., Yuan, Y., Lu, Y. Journal of Mechanical Science and Technology, 2023 https://www.scopus.com/pages/publications/85173744505?origin=scopusAI

54. SIMULATION OPTIMIZATION AND EXPERIMENTAL STUDY OF THE AIR DISTRIBUTION CHAMBER STRUCTURE OF STRAW-BASED NUTRIENT SEEDING-GROWING BOWL TRAY MICROWAVE–HOT-AIR COUPLING DRYERS Li, H., Hu, Y., Qi, L., Yu, H. Engenharia Agricola, 2022 https://www.scopus.com/pages/publications/85129709913?origin=scopusAI

55. An improved infrared combined hot air dryer design and effective drying strategy analysis for sweet potato Jiang, D.-L., Shirkole, S.S., Ju, H.-Y., (...), Xiao, H.-W. LWT, 2025 https://www.scopus.com/pages/publications/85211698677?origin=scopusAI

56. Internet of Things-Based Food Product Drying Oven using Shallow Geothermal Heat Situmorang, V., Pratama, Y., Tambunan, I.H., (...), Sagala, A. ICOSNIKOM 2022 - 2022 IEEE International Conference of Computer Science and Information Technology: Boundary Free: Preparing Indonesia for Metaverse Society, 2022 https://www.scopus.com/pages/publications/85149183918?origin=scopusAI

57. Eco-Friendly halogen-free composite for humidity control: Application in environmental humidity regulation Fu, J., Lv, G., Li, T., (...), Peng, Z. Journal of the Taiwan Institute of Chemical Engineers, 2025

https://www.scopus.com/pages/publications/105002572505?origin=scopusAI

58. Architectural design of hydrophobic bimodal PCL/EC fibrous filters for efficient particulate separation under high-humidity environments Nie, W., Liao, M., Guo, L., (...), Hua, Y. Journal of Membrane Science, 2026 https://www.scopus.com/pages/publications/105046568196?origin=scopusAI

59. Sustainable waste-derived thermal energy storage composites for performance enhancement of solar dehydrators in strawberry drying Arbab, M.S., Rehman, W.U., Fadhillah, F., (...), Shah, M.U.H. Applied Thermal Engineering, 2026 https://www.scopus.com/pages/publications/105024852981?origin=scopusAI